

Information-Theoretic Limits of Finetuning in Quantized LLMs

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- **Duration:** 4 to 6 months.
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Context: LLMs are often viewed as overparameterized systems, yet in practice they are *heavily quantized*, so the number of bits used to represent their parameters is small compared to the information content of their training datasets. This places us in a *bit-constrained* / *underquantized* regime where classical intuitions about capacity, memorization, and generalization may no longer apply. In this project, we aim to adopt an information-theoretic viewpoint and treat quantized models as compressed representations of their training data. In the *finetuning* setting—where model weights are quantized both before and after adaptation—we hypothesize that the number of bits that change during finetuning is closely tied to the *compressibility* of the finetuning dataset, suggesting a deep link between learning dynamics and information-theoretic limits. This connects practical quantization methods (e.g., post-training quantization such as GPTQ [2], quantized finetuning via QLoRA [1], and recent Pareto-optimal quantization trade-offs [4]) with description-length perspectives and MDL principles [3].

Goals: We aim to articulate and probe research paths linking the *adaptation bit-budget* to *dataset compressibility*, and to understand how this relationship drives performance and generalization. We will also seek empirical bit–performance laws (accuracy or loss versus adaptation bits) and characterize Pareto trade-offs among bits, speed, and accuracy [4].

Organisation: The internship will start with a survey of the literature on quantization for LLMs and information-theoretic lenses on learning (GPTQ, QLoRA, Pareto trade-offs, MDL). Moreover, a brief review of core NLP architectures will guide the choice of evaluation targets and datasets. Then, the experimental setup will be defined to assess the proposed approach: a compact, reproducible PYTORCH pipeline to finetune quantized models on lightweight benchmarks and to evaluate statistics. There is also a possibility to pursue this work during a PhD thesis in the lab.

References

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